I HARD CODED (XOR GATE)

```
import numpy as np
# Inputs
x1 = np.array([0, 0,1,1])
x2 = np.array([0,1,0,1])
t = np.array([[0],[1],[1],[0]])
lr = 0.05
epochs = 20000
m = len(x1)
# HARD-CODED WEIGHTS AND BIAS
W1 = np.array([[1.0], -1.0], [-1.0, 1.0]])
b1 = np.array([[0.0, 0.0]])
W2 = np.array([[1.0], [1.0]])
b2 = np.array([[0.0]])
# sigmoid
def sigmoid(x):
    return 1/(1+np.exp(-x))
def sigmoid_derivative(x):
    return x*(1-x)
# Training
for _ in range(epochs):
    for i in range(m):
        x = np.array([[x1[i], x2[i]]])
        # forward pass
        h = sigmoid(np.dot(x, W1) + b1)
        y = sigmoid(np.dot(h, W2) + b2)
        error = t[i] - y
        # Backpropagation
        d_output = error * sigmoid_derivative(y)
        d_hidden = np.dot(d_output, W2.T) *
                   sigmoid_derivative(h)
        # update weights and bias
        W2 += lr * np.dot(h.T, d_output)
        b2 += lr * d_output
        W1 += lr * np.dot(x.T, d_hidden)
        b1 += lr * d_hidden
print("XOR output:")
for i in range(m):    x = np.array([[x1[i], x2[i]]])
```

OUTPUT

```
XOR output:
0
1
1
0
```